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UMC Reports Third Quarter 2021 Results

Strategic focus in 5G, auto, AIoT markets will further strengthen UMC's global leading specialty foundry position

Third Quarter 2021 Overview[1]:

- **Revenue: NT$55.91 billion (US$2.01 billion)**
- **Gross margin: 36.8%; Operating margin: 27.1%**
- **Revenue from 22/28nm: 19%**
- **Capacity utilization rate: 100%+**
- **Net income attributable to shareholders of the parent: NT$17.46 billion (US$627 million)**
- **Earnings per share: NT$1.43; earnings per ADS: US$0.257**

Taipei, Taiwan, ROC – October 27, 2021 – United Microelectronics Corporation (NYSE: UMC; TWSE: 2303) ("UMC" or "The Company"), a leading global semiconductor foundry, today announced its consolidated operating results for the third quarter of 2021.

Third quarter consolidated revenue was NT$55.91 billion, increasing 9.8% QoQ from NT$50.91 billion in 2Q21. Compared to a year ago, 3Q21 revenue grew 24.6% YoY from NT$44.87 billion in 3Q20. Consolidated gross margin for 3Q21 was 36.8%. Net income attributable to the shareholders of the parent was NT$17.46 billion, with earnings per ordinary share of NT$1.43.

Jason Wang, UMC co-president, said, "In the third quarter, we continued to experience robust chip demand across computing, consumer and communication end segments. Higher 12" wafer shipments in the quarter reflected ongoing product mix enhancements and partially contributed to the lift in blended ASP. Overall wafer shipments grew 2.6% QoQ to 2.50 million 8-inch equivalents. Revenue from 28nm technologies continued to rise while business engagements in 22nm have led to a growing number of customers' tape outs across wireless, display, and IoT markets, further

[1]Unless otherwise stated, all financial figures discussed in this announcement are prepared in accordance with TIFRSs recognized by Financial Supervisory Commission in the ROC, which is different from IFRSs issued by the International Accounting Standards Board. They represent comparisons among the three-month period ending September 30, 2021, the three-month period ending June 30, 2021, and the equivalent three-month period that ended September 30, 2020. For all 3Q21 results, New Taiwan Dollar (NT$) amounts have been converted into U.S. Dollars at the September 30, 2021 exchange rate of NT$ 27.84 per U.S. Dollar.

diversifying our product pipeline."

Co-president Wang continued, "Looking into the fourth quarter, we anticipate wafer shipments and the ASP trend will remain firm. Capacity utilization across 8" and 12" facilities will continue to remain fully loaded, as gross margin continues to exhibit upward momentum, thanks to our team's efforts in optimizing capacity productivity and product mix. The current business cycle provides an opportune time for UMC to strengthen customer relationships, along with our technology competitiveness and the incremental capacity growth to elevate our market position. Our focus on growing our comprehensive logic and specialty technology portfolio has been welcomed by our customers and we continue to broaden our product range to fulfill their needs. With the P5 & P6 expansion projects underway at our flagship 12A in Tainan, given the strong demand from our customers, we are well positioned to grow and capture additional market share in 2022."

Co-president Wang added, "In addition, the company continue to make strides towards a greener future. Earlier this month, UMC was honored to receive the Green Chemistry Application and Innovation Award from Taiwan's Environmental Protection Agency. The award recognizes our efforts to introduce chemical substitutes that minimize impact to the environment and the health of our employees. At our Outstanding Supplier Awards this year, UMC also took the opportunity to reiterate our commitment to achieve net zero carbon emissions by 2050 and to invite suppliers to work with us to build a low carbon supply chain. As a key semiconductor player, UMC understands that we have a responsibility to proactively respond to climate change and to promote sustainable practices in our industry. Together with our upstream and downstream partners, we will continue working towards our net zero carbon emissions target."

Summary of Operating Results

Operating Results					
(Amount: NT$ million)	3Q21	2Q21	QoQ % change	3Q20	YoY % change
Operating Revenues	55,907	50,908	9.8	44,870	24.6
Gross Profit	20,544	15,908	29.1	9,769	110.3
Operating Expenses	(6,636)	(6,201)	7.0	(5,508)	20.5
Net Other Operating Income and Expenses	1,227	1,606	(23.6)	2,872	(57.3)
Operating Income	15,135	11,313	33.8	7,133	112.2
Net Non-Operating Income and Expenses	4,317	1,881	129.5	2,074	108.2
Net Income Attributable to Shareholders of the Parent	17,460	11,943	46.2	9,106	91.7
EPS (NT$ per share)	1.43	0.98		0.75	
(US$ per ADS)	0.257	0.176		0.135	

Operating revenues in 3Q21 increased 9.8% to NT$55.91 billion which included higher 12” wafer shipments and the optimization in product mix. Revenue contribution from 40nm and below technologies was 37%. Gross profit grew 29.1% QoQ to NT$20.54 billion, or 36.8% of revenue. Operating expenses increased 7.0% to NT$6.64 billion. Net other operating income declined to NT$1.23 billion. Net non-operating income increased to NT$4.32 billion. Net income attributable to shareholders of the parent increased to NT$17.46 billion.

Earnings per ordinary share for the quarter was NT$1.43. Earnings per ADS was US$0.257. The basic weighted average number of outstanding shares in 3Q21 was 12,206,292,756, compared with 12,206,292,756 shares in 2Q21 and 12,107,651,452 shares in 3Q20. The diluted weighted average number of outstanding shares was 12,411,100,649 in 3Q21, compared with 12,382,592,798 shares in 2Q21 and 12,179,561,492 shares in 3Q20. The fully diluted shares counted on September 30, 2021 were approximately 12,427,179,000.

Detailed Financials Section

Operating revenues increased to NT$55.91 billion. COGS went up 1.0 % QoQ to NT$35.36 billion, as the 2.8% sequential decline in depreciation was offset by the 2.6% increase in other manufacturing costs mainly from higher wafer shipments in 3Q21. Gross profit increased 29.1% QoQ to NT$20.54 billion, reflecting the uptick in overall average selling price across 8" and 12" wafer shipments. Operating expenses increased 7.0% QoQ to NT$6.64 billion, as G&A grew 11.5% sequentially to NT$2.12 billion while Sales & Marketing increased 7.2% QoQ to NT$1.21 billion. R&D increased 4.3% to NT$3.30 billion, representing 5.91% of revenue. Net other operating income was NT$1.23 billion. In 3Q21, operating income grew 33.8% QoQ to NT$15.14 billion.

COGS & Expenses

(Amount: NT$ million)	3Q21	2Q21	QoQ % change	3Q20	YoY % change
Operating Revenues	55,907	50,908	9.8	44,870	24.6
COGS	(35,363)	(35,000)	1.0	(35,101)	0.7
Depreciation	(9,900)	(10,187)	(2.8)	(10,911)	(9.3)
Other Mfg. Costs	(25,463)	(24,813)	2.6	(24,190)	5.3
Gross Profit	20,544	15,908	29.1	9,769	110.3
Gross Margin (%)	36.8%	31.3%		21.8%	
Operating Expenses	(6,636)	(6,201)	7.0	(5,508)	20.5
G&A	(2,119)	(1,901)	11.5	(1,614)	31.2
Sales & Marketing	(1,212)	(1,131)	7.2	(1,009)	20.1
R&D	(3,303)	(3,168)	4.3	(3,314)	(0.3)
Expected Credit Impairment Gain (Loss)	(2)	(1)	46.7	429	-
Net Other Operating Income & Expenses	1,227	1,606	(23.6)	2,872	(57.3)
Operating Income	15,135	11,313	33.8	7,133	112.2

Net non-operating income in 3Q21 was NT$4.32 billion, primarily reflecting NT$4.53 billion in net investment gain and a NT$164 million in exchange gain, offset by a NT$367 million in net interest expense.

Non-Operating Income and Expenses

(Amount: NT$ million)	3Q21	2Q21	3Q20
Non-Operating Income and Expenses	4,317	1,881	2,074
Net Interest Income and Expenses	(367)	(310)	(314)
Net Investment Gain and Loss	4,534	2,276	3,944
Exchange Gain and Loss	164	(84)	259
Other Gain and Loss	(14)	(1)	(1,815)

In 3Q21, cash inflow from operating activities was NT$22.13 billion. Cash outflow from investing activities amounted to NT$23.22 billion, which included NT$17.88 billion in capital expenditure, resulting in free cash flow of NT$4.25 billion. Cash outflow from financing activities was NT$9.69 billion, mainly from a NT$19.87 billion in the distribution of cash dividend and a NT$4.51 billion in payment in bank loans, offset by a NT$11.16 billion in bond issuance and a NT$3.74 billion payment as capacity deposit for 12A P6. Net cash outflow in 3Q21 was NT$10.90 billion. Over the next 12 months, the company expects to repay NT$6.75 billion in bank loans.

Cash Flow Summary

(Amount: NT$ million)	For the 3-Month Period Ended Sep. 30, 2021	For the 3-Month Period Ended Jun. 30, 2021
Cash Flow from Operating Activities	22,125	23,123
Net income before tax	19,452	13,194
Depreciation & Amortization	11,754	11,671
Share of profit of associates and joint ventures	(3,356)	(881)
Income tax paid	(1,365)	(300)
Changes in working capital & others	(4,360)	(561)
Cash Flow from Investing Activities	(23,222)	(18,482)
Acquisition of PP&E	(17,457)	(8,520)
Acquisition of intangible assets	(371)	(835)
Increase in other financial assets	(5,677)	(10,893)
Others	283	1,766
Cash Flow from Financing Activities	(9,694)	13,238
Bank loans	(4,510)	(1,384)
Bonds Issued	11,161	9,600
Redemption of bonds	-	(2,000)
Increase in deposits-in	3,738	7,205
Cash dividends	(19,870)	-
Others	(213)	(183)
Effect of Exchange Rate	(104)	(1,164)
Net Cash Flow	(10,895)	16,715
Beginning balance	124,000	107,285
Ending balance	113,105	124,000

Cash and cash equivalents decreased to NT$113.11 billion. Days of inventory increased by a day to 59 days.

Current Assets			
(Amount: NT$ billion)	3Q21	2Q21	3Q20
Cash and Cash Equivalents	113.11	124.00	98.84
Notes & Accounts Receivable	32.99	30.11	26.96
Days Sales Outstanding	51	53	55
Inventories, net	23.15	22.44	22.86
Days of Inventory	59	58	60
Total Current Assets	211.60	207.83	163.48

Current liabilities decreased to NT$84.55 billion, which included a NT$24.26 billion in short-term credit/bonds. Long-term credit/bonds increased slightly to NT$51.09 billion. Total liabilities declined to NT$171.19 billion, leading to a debt to equity ratio of 66%.

Liabilities			
(Amount: NT$ billion)	3Q21	2Q21	3Q20
Total Current Liabilities	84.55	85.00	59.15
Notes & Accounts Payable	8.26	8.42	7.70
Short-Term Credit / Bonds	24.26	19.65	16.40
Payables on Equipment	6.72	6.67	7.38
Dividends Payable	-	19.88	-
Other	45.31	30.38	27.67
Long-Term Credit / Bonds	51.09	50.97	49.46
Long-Term Investment Liabilities	8.14	20.61	20.14
Total Liabilities	171.19	181.49	147.33
Debt to Equity	66%	76%	67%

Analysis of Revenue[2]

Revenue from Asia-Pacific increased to 65% as business from North America remained at 22% of sales. Business from Europe was 7% while contribution from Japan declined to 6%.

Revenue Breakdown by Region

Region	3Q21	2Q21	1Q21	4Q20	3Q20
North America	22%	22%	23%	29%	30%
Asia Pacific	65%	63%	63%	61%	57%
Europe	7%	8%	8%	5%	6%
Japan	6%	7%	6%	5%	7%

Revenue contribution from 22/28nm was 19% of the wafer revenue, while 40nm contribution stayed at 18% of sales.

Revenue Breakdown by Geometry

Geometry	3Q21	2Q21	1Q21	4Q20	3Q20
14nm and below	0%	0%	0%	0%	0%
14nm<x<=28nm	19%	20%	20%	18%	14%
28nm<x<=40nm	18%	18%	20%	22%	23%
40nm<x<=65nm	19%	19%	18%	18%	19%
65nm<x<=90nm	8%	9%	8%	8%	10%
90nm<x<=0.13um	12%	11%	11%	11%	11%
0.13um<x<=0.18um	13%	13%	13%	13%	13%
0.18um<x<=0.35um	8%	8%	8%	8%	8%
0.5um and above	3%	2%	2%	2%	2%

Revenue from fabless customers accounted for 86% of revenue.

Revenue Breakdown by Customer Type

Customer Type	3Q21	2Q21	1Q21	4Q20	3Q20
Fabless	86%	84%	86%	87%	88%
IDM	14%	16%	14%	13%	12%

Revenue from the communication segment declined to 46%, while business from computer applications remained at 17%. Business from consumer applications increased to 27% as other segments accounted for 10% of revenue.

Revenue Breakdown by Application [(1)]

Application	3Q21	2Q21	1Q21	4Q20	3Q20
Computer	17%	17%	16%	16%	13%
Communication	46%	47%	46%	49%	54%
Consumer	27%	26%	27%	25%	24%
Others	10%	10%	11%	10%	9%

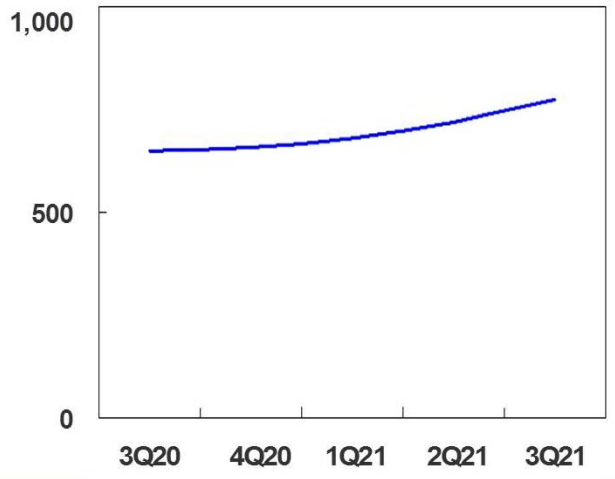
(1) ***Computer*** consists of ICs such as CPU, GPU, HDD controllers, DVD/CD-RW control ICs, PC chipset, audio codec, keyboard controller, monitor scaler, USB, I/O chipset, WLAN. ***Communication*** consists of handset components, broadband, bluetooth, Ethernet, LAN, DSP, etc. ***Consumer*** consists of ICs used for DVD players, DTV, STB, MP3/MP4, flash controller, game consoles, DSC, smart cards, toys, etc.

[2] Revenue in this section represents wafer sales

Blended ASP Trend

Blended average selling price (ASP) grew in 3Q21.

Unit: USD



Shipment and Utilization Rate[3]

In 3Q21, wafer shipments increased 2.6% QoQ to 2,503K, while quarterly capacity grew to 2,383K. Overall utilization rate in 3Q21 exceeded 100%.

Wafer Shipments

	3Q21	2Q21	1Q21	4Q20	3Q20
Wafer Shipments (8" K equivalents)	2,503	2,440	2,372	2,293	2,254

Quarterly Capacity Utilization Rate

	3Q21	2Q21	1Q21	4Q20	3Q20
Utilization Rate	100%+	100%+	100%	99%	97%
Total Capacity (8" K equivalents)	2,383	2,370	2,280	2,311	2,308

[3] Utilization Rate = Quarterly Wafer Out / Quarterly Capacity

Capacity[4]

Total capacity in the third quarter grew by 0.5% QoQ to 2,383K 8-inch equivalent wafers. Capacity is expected to slightly increase in the fourth quarter to 2,419K 8-inch equivalent wafers, mainly reflecting the capacity expansion at Fab 12i, 12M, 12X and 8N.

Annual Capacity in *thousands of wafers*

FAB		Geometry (um)	2020	2019	2018	2017
WTK	6"	3.5 – 0.45	371	370	396	422
8A	8"	0.5 – 0.25	802	825	825	825
8C	8"	0.35 – 0.11	452	436	383	357
8D	8"	0.18 – 0.09	371	359	347	341
8E	8"	0.5 – 0.15	449	426	418	418
8F	8"	0.18 – 0.11	485	434	431	417
8S	8"	0.18 – 0.11	373	372	372	347
8N	8"	0.5 – 0.11	917	831	771	753
12A	12"	0.13 – 0.014	1044	997	997	970
12i	12"	0.13 – 0.040	628	595	555	537
12X	12"	0.040 – 0.028	217	203	183	97
12M	12"	0.090 – 0.040	391	98	-	-
Total[(1)]			9,188	8,148	7,673	7,304
YoY Growth Rate			13%	6%	5%	5%

Quarterly Capacity in *thousands of wafers*

FAB	4Q21E	3Q21	2Q21	1Q21
WTK	81	80	84	84
8A	190	190	190	186
8C	115	115	115	113
8D	95	95	95	94
8E	115	115	115	113
8F	137	137	122	120
8S	102	102	102	101
8N	232	230	230	226
12A	271	271	271	257
12i	164	160	160	157
12X	78	74	74	59
12M	104	98	98	96
Total	**2,419**	**2,383**	**2,370**	**2,280**

[(1)] One 6-inch wafer is converted into 0.5625 ($6^2/8^2$) 8-inch equivalent wafer; one 12-inch wafer is converted into 2.25 ($12^2/8^2$) 8-inch equivalent wafers. Total capacity figures are expressed in 8-inch equivalent wafers.

CAPEX

CAPEX spending in 3Q21 was US$642 million. Full year 2021 CAPEX is budgeted at US$2.3 billion, which includes the company's collaborative Fab 12A P6 expansion plan with customers.

Capital Expenditure by Year - in US$ billion

Year	2020	2019	2018	2017	2016
CAPEX	$ 1.0	$ 0.6	$ 0.7	$ 1.4	$ 2.8

2021 CAPEX Plan

8"	12"	Total
15%	85%	US$2.3 billion

[4] Estimated capacity numbers are based on *calculated maximum output* rather than *designed capacity*. The actual capacity numbers may differ depending upon equipment delivery schedules, pace of migration to more advanced process technologies, and other factors affecting production ramp-up.

Fourth Quarter 2021 Outlook & Guidance

Quarter-over-Quarter Guidance:

- Wafer Shipments: To increase by 1-2%
- ASP in USD: To increase by 1-2%
- Gross Profit Margin: To be in the high-30% range
- Capacity Utilization: 100%
- 2021 CAPEX: US$2.3 billion

Recent Developments / Announcements

Sep 3, 2021	UMC and Chipbond to Establish Strategic Cooperation Through the Exchange of New Share Issuance
Jun 1, 2021	UMC Pledges Net Zero Carbon Emissions by 2050

Please visit UMC's website for further details regarding the above announcements

Conference Call / Webcast Announcement

Wednesday, October 27, 2021

Time: 5:00 PM (Taipei) / 5:00 AM (New York) / 10:00 AM (London)

Dial-in numbers and Access Codes:

USA Toll Free:	1-866 836-0101
Taiwan Number:	02-2192-8016
Other Areas:	+886-2-2192-8016

Access Code: UMC

A live webcast and replay of the 3Q21 results announcement will be available at www.umc.com under the “Investors / Events” section.

About UMC

UMC (NYSE: UMC, TWSE: 2303) is a leading global semiconductor foundry. The company provides high quality IC production with a focus on both logic and specialty technologies to serve every major sector of the electronics industry. UMC’s comprehensive technology and manufacturing solutions include logic/RF, embedded high voltage, embedded flash, RFSOI/BCD and IATF-16949 automotive manufacturing certification for all its manufacturing facilities. UMC operates 12 fabs that are strategically located throughout Asia with a maximum capacity of approximately 800,000 8-inch equivalent wafers per month. The company employs approximately 19,500 people worldwide, with offices in Taiwan, China, United States, Europe, Japan, Korea and Singapore. For more information, please visit: http://www.umc.com.

Safe Harbor Statements

This press release contains forward-looking statements within the meaning of Section 27A of the United States Securities Act of 1933, as amended, and Section 21E of the United States Securities Exchange Act of 1934, as amended, and as defined in the United States Private Securities Litigation Reform Act of 1995. These forward-looking statements include, but are not limited to, statements regarding anticipated financial results for the third quarter of 2021; the expected wafer shipment and ASP; the anticipated annual budget; capex strategies; environmental protection goals and water management strategies; impact of foreign currency exchange rates; expected foundry capacities; the ability to obtain new business opportunities; and information under the heading "Third Quarter of 2021 Outlook and Guidance."

These forward-looking statements involve known and unknown risks, uncertainties and other factors that may cause the actual performance, financial condition or results of operations of UMC to be materially different from what is stated or may be implied in such forward-looking statements. Investors are cautioned that actual events and results could differ materially from those statements as a result of a number of factors including, but not limited to: (i) dependence upon the frequent introduction of new services and technologies based on the latest developments in the industry in which UMC operates; (ii) the intensely competitive semiconductor, communications, consumer electronics and computer industries and markets; (iii) the risks associated with international business activities; (iv) dependence upon key personnel; (v) general economic and political conditions; (vi) possible disruptions in commercial activities caused by natural and human-induced events and disasters, including natural disasters, terrorist activity, armed conflict and highly contagious diseases; (vii) reduced end-user purchases relative to expectations and orders; and (viii) fluctuations in foreign currency exchange rates. Further information regarding these and other risk factors is included in UMC's filings with the United States Securities and Exchange Commission, including its Annual Report on Form 20-F. All information provided in this release is as of the date of this release and are based on assumptions that UMC believes to be reasonable as of this date, and UMC does not undertake any obligation to update any forward-looking statement as a result of new information, future events or otherwise, except as required under applicable law.

The financial statements included in this release are prepared and published in accordance with Taiwan International Financial Reporting Standards, or TIFRSs, recognized by the Financial Supervisory Commission in the ROC, which is different from International Financial Reporting Standards, or IFRSs, issued by the International Accounting Standards Board. Investors are cautioned that there may be significant differences between TIFRSs and IFRSs. In addition, TIFRSs and IFRSs differ in certain significant respects from generally accepted accounting principles in the ROC and generally accepted accounting principles in the United States.

- FINANCIAL TABLES TO FOLLOW -